

Referee report on “Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds”

Summary

The paper examines the economic determinants of bond-stock betas within a standard New Keynesian (NK) framework. The modeling section is divided into two parts.

(P1) The first part derives key model state variables, i.e., output gap x_t , wage inflation π_t^w , interest rate i_t , from a log-linearized New Keynesian (NK) model:

$$x_t = f^x E_t x_{t+1} + \rho^x x_{t-1} - \psi(i_t - \overbrace{[E_t \pi_{t+1}^w - (1 - \phi)x_t]}^{E_t \pi_{t+1}}) + v_{x,t} \quad (\text{Euler equation})$$

$$\pi_t^w = f^\pi E_t \pi_{t+1}^w + \rho^\pi \pi_{t-1}^w + \kappa x_t + v_{\pi,t} \quad (\text{Phillips curve})$$

$$i_t = \rho^i i_{t-1} + (1 - \rho^i) \left(\gamma^x x_t + \gamma^\pi \underbrace{[\pi_t^w - (1 - \phi)x_{t-1}]}_{\pi_t} \right) + v_{i,t} \quad (\text{Taylor rule})$$

(P2) The second part assumes a relationship between the output gap and consumption growth

$$\Delta c_{t+1} = x_{t+1} - \phi x_t \quad (\text{Consumption growth-output gap relation})$$

and imposes habit formation preferences

$$s_{t+1} = (1 - \theta_0)\bar{s} + \theta_0 s_t + \theta_1 x_t + \theta_2 x_{t-1} + \lambda(s_t) \varepsilon_{c,t+1} \quad (\text{Surplus consumption})$$

to obtain the stochastic discount factor

$$m_{t+1} = \ln \beta - \gamma \Delta s_{t+1} - \gamma \Delta c_{t+1}. \quad (\text{SDF})$$

Together with the key state variables derived earlier, this framework is used to derive nonlinear asset (stocks, real and nominal bonds) prices.

In this sense, the paper’s setup builds on the author’s previous two published papers, [Campbell, Pflueger, and Viceira \(2020\)](#) and [Pflueger and Rinaldi \(2022\)](#), as they both include (P1) and (P2). While the model (P1) in [Campbell, Pflueger, and Viceira \(2020\)](#) is not as micro-founded as the one in the current paper, (P1) in [Pflueger and Rinaldi \(2022\)](#) is very similar, **but with some important changes such as enriching structural shocks to include both demand and supply, and considering adaptive subjective inflation expectations.**

Table 1: Calibration

Taylor rule	[R1]	γ^x	0.125	γ^π	1.35	ρ^i	0.54
	[R2]	γ^x	0.25	γ^π	1.10	ρ^i	0.80
Phillips curve	[R1]	κ	0.0015	ρ^π	0.803	ζ	0.60
	[R2]	κ	0.0015	ρ^π	0.508	ζ	0.0
Stdev of shocks	[R1]	$100\sigma_x$	0.01	$100\sigma_\pi$	0.145	$100\sigma_i$	0.138
	[R2]	$100\sigma_x$	0.59	$100\sigma_\pi$	0.018	$100\sigma_i$	0.018

The model is calibrated separately for two distinct subperiods: [R1] 1979:Q4-2001 and [R2] 2001:Q2-2019. The breakpoint date, as per [Campbell, Pflueger, and Viceira \(2020\)](#), was identified through testing for a break in the inflation-output gap relationship, with no reliance on asset prices.

The calibration of the model for two distinct subperiods sets the current paper apart from [Pflueger and Rinaldi \(2022\)](#), while its unique ability to perform counterfactual exercises assessing the impact of Taylor-rule coefficients and macroeconomic shock volatilities on bond-stock betas amplifies its significance.

In the literature, various channels explain bond-stock comovement, including the inflation-risk narrative proposed by [David and Veronesi \(2013\)](#), [Song \(2017\)](#), [Campbell, Pflueger, and Viceira \(2020\)](#), [Gourio and Ngo \(2020\)](#), and [Fang, Liu, and Roussanov \(2022\)](#); the liquidity, flight-to-safety, and time-varying risk aversion argument presented by [Baele, Bekaert, and Inghelbrecht \(2010\)](#), [Kozak \(2022\)](#), and [Laarits \(2022\)](#); examination of permanent-transitory compositional changes discussed by [Chernov, Lochstoer, and Song \(2021\)](#); and the examination of monetary-fiscal mix regimes explored by [Li, Zha, Zhang, and Zhou \(2022\)](#), among others.

What sets the current paper apart is its remarkable ability to integrate these popular narratives within a [relatively simple unified framework](#). It adeptly incorporates elements such as the inflation-risk narrative through the differentiation of demand versus supply shocks, time-varying risk aversion via habit-formation preferences, adjustments in the persistence of state variables, achieved by calibrating the model for two distinct subperiods, and modifications in the monetary policy rule, all contributing to a comprehensive understanding of bond-stock comovement.

Essential points

Overall, I believe the paper merits publication in the *Journal of Financial Economics*. While I find the current paper already quite clear, I will only raise two relatively minor points.

Firstly, considering that the paper's primary contribution focuses on explaining changing bond-stock comovement, I recommend that the author provide a more precise explanation of her contribution to the literature. Although the author cites most of the papers mentioned above in the paper, many of them are relegated to footnotes. For example, among the more pertinent papers, the author's paper undoubtedly presents advancements beyond Song (2017), which also discussed changes in the monetary policy rule within an endowment economy, and it extends Campbell, Pflueger, and Viceira (2020), where the monetary policy rule isn't specified. None of the production-based papers mentioned above, such as Gourio and Ngo (2020), Fang, Liu, and Roussanov (2022), and Kozak (2022), incorporate changes in the monetary policy rule. However, the work of Li, Zha, Zhang, and Zhou (2022) is somewhat unclear, as it also explores changes in policy rules, encompassing both monetary and fiscal aspects. The distinction may lie in the fact that Li, Zha, Zhang, and Zhou (2022) do not consider changes in volatility related to structural shocks. Clarifying both the precise contribution of this paper and that of others would be beneficial for readers to comprehend its position in the literature.

Secondly, while I generally agree with the author's calibration across two periods, denoted as R1 and R2 in my report, I recommend that the author provide more discussion regarding the existing macro literature on calibration. The author has touched upon this to some extent in footnote 4, where she discussed the shift to a more inertial monetary policy rule with greater emphasis on output, aligning with anecdotal evidence. However, the part concerning dominant supply shocks in the 80s versus demand shocks in the 00s may warrant further discussion. In particular, Table 1 highlights a substantial difference in the magnitudes of volatilities σ_j for $j \in \{\chi, \pi, i\}$ across the two periods. Two notable references, Liu, Waggoner, and Zha (2011) and Bianchi and Ilut (2017), offer diverse volatility estimates within regime-switching New Keynesian models, both of which are highly cited. However, given the intricacies of their respective frameworks, which incorporate a broader array of structural shocks, direct comparisons may be challenging. In light of this complexity, it becomes imperative for the reader to discern whether the current paper's calibration, particularly on (P1), aligns broadly with the trends observed in the macro DSGE literature. Should disparities arise, elucidating the rationale behind such deviations becomes essential. This clarity is paramount, especially considering that subsequent counterfactual analyses pivot critically on the calibration values delineated in Table 1.

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